

Evaluating Carbon Capture Technologies for Climate Change

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Abstract

Fossil fuels remain the dominant source of global energy production, despite contributing to climate change by releasing carbon dioxide (CO₂) in the atmosphere. Carbon capture technologies are a potential solution to this problem, which aim to reduce CO₂ levels by preventing its release at the source or removing it directly from the air. This paper examines underground coal gasification (UCG), oxyfuel combustion, and direct air capture (DAC) to evaluate the most balanced and practical technology. The analysis is based on economic feasibility, efficiency, and environmental impact. While each method has advantages, oxyfuel combustion is deemed the most balanced solution for large-scale industrial settings due to its low cost, high capture rates, and strong environmental impact. That said, UCG has merits regarding cost and efficiency as a sustainable alternative to coal mining, while DAC has the best long-term sustainability potential but is limited by high costs and low efficiency. By comparing these technologies, this paper highlights the importance of finding the best carbon capture method for sustainability and industrial purposes.

Keywords: Carbon capture, climate change, underground coal gasification, oxyfuel combustion, direct air capture

In today's rapidly advancing world, climate change has become a pressing global issue due to increasing industrialization and societal development. The burning of fossil fuels releases CO₂ into the atmosphere, intensifying the greenhouse effect and increasing global temperatures. A major contributor to this problem is anthropogenic CO₂ emissions, primarily from the increasing demand for energy and manufacturing in industrial settings. In response, carbon capture and storage technologies have emerged as a viable option to mitigate CO₂ emissions and promote environmental sustainability (Rubin et al., 2012). This research paper explores these solutions, analyzing three carbon capture methods: underground coal gasification, oxyfuel combustion, and direct air capture. Against the criteria of economic feasibility, efficiency, and overall environmental impact, this paper argues that oxyfuel combustion presents the most balanced and effective solution for large-scale industrial applications.

Background

The modern world relies heavily on fossil fuels for energy production, as they account for over 85% of global energy consumption, stimulating interest in carbon capture technology as a solution (Rubin et al., 2012). While transitioning to renewable energy can be a long-term solution, these efforts are time-consuming and expensive, leading to insufficient carbon reductions in the short term. Carbon capture technologies offer a practical solution by allowing the continued use of fossil fuels while simultaneously reducing CO₂ emissions.

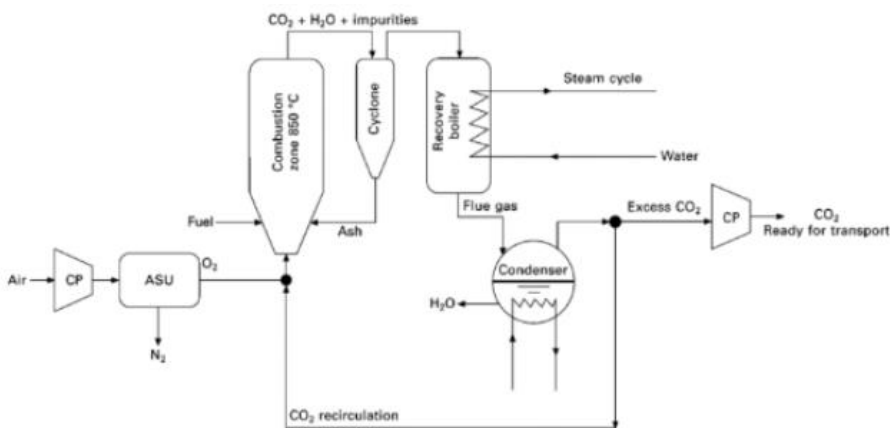
Beyond maintaining current energy demands, carbon capture plays a crucial role in achieving sustainability goals necessary to combat climate change. According to The United Nations Framework Convention on Climate Change (2015), the Paris Agreement aims to "bring global temperatures to below 2°C" and should ultimately "limit temperatures to 1.5°C". To achieve this target, cumulative global CO₂ emissions must not exceed 320 GtCO₂ or 740 GtCO₂, with a 66% and 33% chance for success, respectively (Gambhir & Tavoni, 2019). However, with current anthropogenic CO₂ emissions reaching 40 GtCO₂ with annual increases of 1%, these limits will be exceeded soon (Gambhir & Tavoni, 2019). Consequently, researchers have been heavily involved in developing carbon capture technologies to achieve sustainability goals.

Overview of Carbon Capture Technologies

Oxyfuel Combustion

Oxyfuel combustion is a widely used carbon capture technology in industrial settings. Unlike conventional combustion methods that use air and nitrogen as reactants, oxyfuel combustion uses near-pure oxygen to produce a concentrated CO₂ stream (Mathieu, 2010). The process for oxyfuel combustion is depicted in Figure 1. First, air separation units split incoming air into streams of nitrogen and oxygen, yielding 95-99 % pure oxygen streams that are used in the combustion process to generate flue gas (Rubin et al., 2012). Since pure oxygen combustion reaches high temperatures of 2500°C, the temperature in the combustion zone is controlled by recycling flue gas and excess oxygen out of circulation (Mathieu, 2010). Finally, water vapor is removed from the flue gas via condensation, resulting in a CO₂ stream with purities ranging from 80-98%, which can then be stored and transported for other purposes (Mathieu, 2010).

Figure 1: Oxyfuel Combustion Process (Mathieu, 2010)

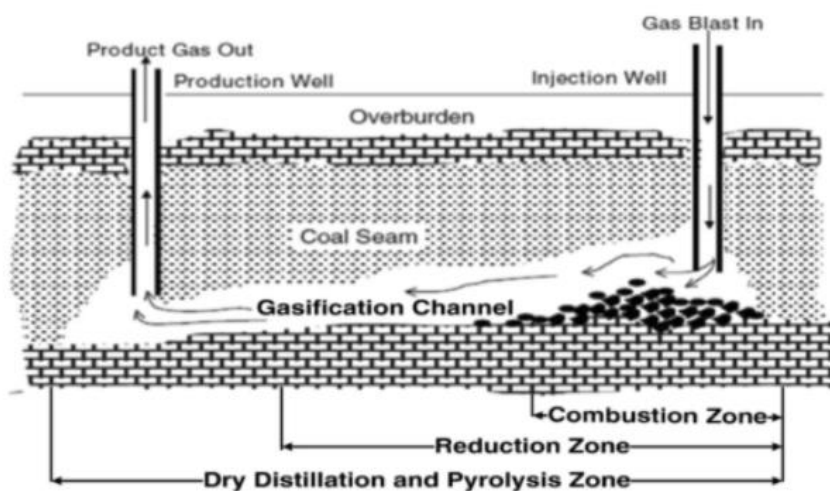


Gasification

Gasification is a carbon capture technology that converts carbonaceous fuels into syngas, which can then be used in carbon sequestration (Man et al., 2014). Underground coal gasification (UCG) is a specific gasification method occurring in natural geological sites containing unmined coal (Self et al., 2012). As illustrated in Figure 2, the UCG process begins

with drilling injection and production wells into the ground where the unmined coal resides (Self et al., 2012). Once the coal is ignited, compressed gasification agents, such as air or steam, flow through the injection well and react with the coal to produce syngas (Bell et al., 2010, p. 101). The Rectisol method, a process that removes acidic gases from gas streams, is then used to extract CO₂ from the resulting syngas (Man et al., 2014). Crude syngas passes through a water scrubber to remove chemical residues, after which a mixture of acid gas and methanol separates the CO₂ from the other elements (Man et al., 2014). This method captures over 90% of the CO₂, which is then stored for later use (Man et al., 2014). UCG is an effective carbon capture technology that reduces emissions associated with coal mining, making it an environmentally friendlier alternative.

Figure 2: Underground Coal Gasification Process (Self et al., 2012)



Direct Air Capture

Direct air capture (DAC) is a carbon capture technology that differs significantly from alternative methods. While UCG and oxyfuel combustion limit CO₂ emissions at the source, DAC removes CO₂ directly from the atmosphere, enabling net negative emissions (The International Energy Agency [IEA], 2022, p. 49). Despite CO₂ occurring in the atmosphere at just 400 ppm, DAC can still remove significant quantities of CO₂ by exposing large volumes of air to chemical sorbents (Gambhir & Tavoni, 2019). The DAC process consists of two main stages: capture and regeneration (Azarabadi & Lackner, 2019). During the capture stage,

atmospheric CO₂ binds to a chemical sorbent, and in the regeneration stage, the CO₂ is separated for sequestration (Azarabadi & Lackner, 2019). During the capture stage, both liquid and solid sorbents can be used, with liquid-based sorbents forming carbonate solutions while solid amine-based sorbents adsorb CO₂, which is subsequently extracted under low temperatures (Gambhir & Tavoni, 2019). Despite its potential, DAC remains an emerging technology, which is why further research must be conducted to determine its effectiveness in carbon removal.

Key Criteria for Comparison

Three criteria will be used to determine the most balanced and practical technology: economic feasibility, efficiency, and environmental impact. Regarding economic feasibility, this research paper will assess the initial investment required, including capital costs and operational expenses. Additionally, the capture cost per ton of CO₂ will be examined as well. Regarding efficiency, the focus will be on the CO₂ capture rate, which is the amount of CO₂ captured per unit of energy. Furthermore, the impact of implementing carbon capture on overall plant efficiency will be evaluated, as some methods may reduce energy output and efficiency. Finally, for environmental impact, this paper will assess the effectiveness in reducing CO₂ emissions and evaluate the long-term sustainability of each technology. Moreover, the analysis will consider whether the addition of certain methods poses environmental risks to surrounding ecosystems.

Analysis

Economic Feasibility

Oxyfuel Combustion

Oxyfuel combustion involves high operational and capital costs due to the need for pure oxygen. The annual cost of operating an oxyfuel combustion power plant is 52.83 M\$, while fuel costs for pure oxygen are 226.33 M\$ (Seo et al., 2021). Additionally, to build a 600MW oxyfuel combustion power plant, an average initial investment of 1.664 B\$ is required (Seo et al., 2021). Despite the high capital and maintenance costs, this technology is inexpensive in terms of capture costs at around 35 \$/tCO₂ (García-Luna et al., 2022). As a result, from an economic

standpoint, the costs of oxyfuel combustion are balanced, offsetting a large initial investment with cheaper capture costs.

Gasification

Regarding capital and investment costs, gasification is a cost-effective option. The life cycle cost of a UCG plant is 61.8 \$/MWh, but integrating the Rectisol method increases this by 18.96%, bringing the total to 73.52 \$/MWh (Feng et al., 2024). In terms of investments, the initial capital cost for a 100 MW gasification project ranges from 210 M\$ to 246 M\$, with low capture costs of around 27 - 28 \$/tCO₂ (Feng et al., 2024). These factors indicate that UCG is a less costly technology than oxyfuel combustion, displaying low capture and capital costs.

Direct Air Capture

DAC is the most expensive carbon capture technology with uncertain and high costs, as it is still in its emerging stages. Additionally, the IEA (2022) believes that it is much more costly to capture CO₂ from the air as it is far more diluted in the atmosphere (p. 27). For liquid and solid sorbents, capture costs range from 95 - 230 \$/tCO₂ and 100 - 600 \$/tCO₂, respectively, while initial capital costs are 65 M\$ and 125 M\$, respectively (IEA, 2022, p. 27; Ozkan et al., 2022). Moreover, for both systems, the maintenance costs range from 50 - 100 \$/tCO₂ (Ozkan et al., 2022). This shows that from a cost perspective, liquid-based DAC is preferable to solid-based DAC, but the costs for both are more uncertain and higher than oxyfuel combustion or UCG.

Efficiency

Oxyfuel Combustion

Oxyfuel combustion is a highly efficient carbon capture method, with capture rates up to 90% (Talei et al., 2024). However, like with most carbon capture technologies, implementing oxyfuel combustion will inevitably decrease overall plant efficiency due to the energy demands to form pure oxygen. Without oxyfuel combustion, net plants yield an average efficiency of 40%, dropping to 32% after adding oxyfuel combustion (García-Luna et al., 2022).

Gasification

Similar to oxyfuel combustion, UCG has high carbon capture efficiency, with a capture rate of 90% (Man et al., 2014). However, this value is limited due to syngas leaks and the high energy consumption of the Rectisol method (Man et al., 2014). Additionally, the implementation of CO₂ capture also decreases the net electrical efficiency of the power plant from 36% to 30.4% (Feng et al., 2021). This 5.6% decrease is lower than the 8% decrease from oxyfuel combustion, indicating UCG is slightly more efficient than oxyfuel combustion.

Direct Air Capture

Since DAC takes CO₂ directly from the air, a drawback regarding efficiency is its dependence on the humidity of the atmosphere. At temperatures above 17 °C and 90% humidity, capture rates are around 75%, but in colder or drier conditions, capture rates can be as low as 55% (An et al., 2022). Despite this limitation, DAC has the advantage of operating independently, thereby not needing to be integrated into existing plants, avoiding any reduction in overall plant efficiency. This is a strong point, though its low capture rates make this method the least efficient of its alternatives.

Environmental Preservation

Oxyfuel Combustion

Oxyfuel combustion is an effective carbon capture technology that significantly reduces air pollutants with few environmental drawbacks. Since it separates oxygen from air, it allows harmful gases such as NO_x and SO_x to be significantly reduced compared to conventional combustion methods, with removal efficiencies of 94-98% and 99.8%, respectively (Senior et al., 2013). Regarding long-term usage, oxyfuel technology is a low-risk, high-reward technology since its processes are well-proven and commercially available, making it a reliable option for large-scale implementation (Mathieu, 2010). Consequently, its environmental benefits make oxyfuel combustion a highly sustainable option for CO₂ mitigation.

Gasification

UCG presents many environmental benefits compared to traditional coal mining, though it still has some drawbacks when compared to its alternatives. Unlike conventional coal mining, UCG removes many environmental risks, leaving coal ash and sulfur oxides underground, thereby holding merits for long-term sustainability (Feng et al., 2021). Furthermore, UCG decreases the plant's lifetime carbon footprint by 28%, though the amount of carbon emitted in the air is still relatively high at 69 m²/t of syngas (Man et al., 2014). However, the most significant environmental risk associated with UCG is groundwater contamination. Since the residues from UCG are left underground, several compounds can contaminate the groundwater, which negatively impacts the surrounding ecosystem (Bell et al., 2010, p. 107). Despite its advantages, the potential for groundwater contamination is a concern that must be addressed to establish UCG as a viable solution.

Direct Air Capture

DAC is the most promising carbon capture technology due to its ability to take CO₂ directly from the air, though its current impacts are limited. Current technologies for DAC show that modern plants can capture 1MtCO₂ per year, a small amount compared to the 40 GtCO₂ emitted annually (Gambhir & Tavoni, 2019). However, the IEA (2022) predicts that by 2040, DAC will capture over 620 MtCO₂, increasing to 1000 MtCO₂ by 2050, thereby being a strong indicator of its long-term sustainability potential (p. 15). Despite this, DAC technology requires a considerable amount of water to run efficiently. In hot and dry conditions, DAC plants can consume 19.8 tH₂O/tCO₂, posing challenges in regions where water is scarce and limiting DAC's overall impact (An et al., 2022).

To clearly compare each method, Table 1 summarizes this paper's key findings based on economic feasibility, efficiency, and environmental impact.

Table 1: Comparative Summary of Carbon Capture Technologies

	Oxyfuel Combustion	UCG	DAC
Economic Feasibility	High capital investment of 1.664B\$, but low capture costs of 35\$/tCO ₂ (Seo et al., 2021; García-Luna et al., 2022).	Lower capital and capture costs of 210 - 246M\$ and 27-28\$/tCO ₂ , respectively (Feng et al., 2024).	Very high capture and operational costs of 95 – 600\$/tCO ₂ and 50 - 100 \$/tCO ₂ , respectively (IEA, 2022, p. 27; Özkan et al., 2022).

Efficiency	High capture rates of 90%, with an 8% drop in net efficiency (Talei et al., 2024; García-Luna et al., 2022).	High captures rates of 90%, with a 5.6% drop in net efficiency (Man et al., 2014; Feng et al., 2021).	Capture rates of 55-75% depending on humidity; DAC does not reduce plant efficiency (An et al., 2022).
Environmental Impact	Low NOx and SOx emissions, utilizing proven technology (Senior et al., 2013; Mathieu, 2010).	Greener than traditional coal mining, though contains risks of groundwater contamination (Bell et al., 2010, p. 107).	Strong long-term potential, with limited current impact and high-water usage (An et al., 2022)

Conclusion

This paper examined three carbon capture methods — oxyfuel combustion, direct air capture, and underground coal gasification — and found that oxy-fuel combustion is the most balanced solution based on the criteria of economic feasibility, efficiency, and environmental preservation. As shown in Table 1, oxyfuel combustion presents high CO₂ capture efficiency and cost-effectiveness for large-scale industrial applications, while UCG provides high capture rates and low costs, but still has drawbacks regarding environmental preservation. Conversely, DAC has the most potential for sustainability but is currently limited by high costs and energy demands. Consequently, significant advancements in cost and scalability are required before it can be widely implemented. To expand further development on this topic, additional research is required to examine the integration of carbon capture technologies into existing industrial plants to improve their impact on the environment.

Final Recommendation

Given the unique strengths and limitations of each method, an approach that utilizes multiple technologies may be the most effective strategy for achieving climate goals. Based on current technology, the widespread implementation of additional oxyfuel combustion plants would reduce the amount of CO₂ emitted into the atmosphere while considering costs, efficiency, and environmental impact. Additionally, since UCG gasification can act as a sustainable alternative to coal mining, it is recommended that more UCG plants be implemented for environmental preservation. Finally, further research and development of DAC is strongly recommended, as its long-term sustainability potential would be important in the future.

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